INCH POUND

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SUPERSEDING
MIL-M-70742 (AR)
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MILITARY SPECIFICATION

MOUNT, TELESCOPE AND QUADRANT: M187 AND M187A1

This specification is approved for use by the U.S. Army Armament Research, Development and Engineering Center (ARDEC) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1. This specification covers performance requirements, quality assurance provisions and packaging data relative to a telescope mount designed primarily to provide adjustments in pitch and to compensate for angles of cant for indirect fire.

1.2 Classification. The Telescope and Quadrant Mount is primarily intended to support a panoramic telescope on the Light Howitzer and are the following types:

1.2.1 M187. This Telescope Mount is intended to support the Light Howitzer M119A1, and uses tritium.

1.2.2 M187A1. This Telescope Mount is intended to support the Light Howitzer M119 Series, and does not use tritium.

2. APPLICABLE DOCUMENTS.

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements of documents cited in sections 3 or 4 of this specification, whether or not documents are listed.

Comments, suggestions, or questions on this document should be addressed to: Commander, US Army ARDEC, ATTN: AMSRD-AAR-AIS-SS, Picatinny, New Jersey 07806-5000 or emailed to ardec-stdzn@pica.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at http://assist.daps.dla.mil.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-F-13926	-	Fire Control Material; General Specification Governing the Manufacture, and Inspection
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-129	-	Military Marking for Shipment and Storage
MIL-STD-1916	-	DOD Preferred Methods for Acceptance of Product
MIL-STD-2073-1	-	Standard Practice for Military Packaging

(Copies of federal and military specifications, standards, and handbooks are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC) DRAWINGS

5549108	-	Collimator
8565556	-	Pressure Tester
8658940	-	Gunners Quadrant, Calibrated (Inspection Aid)
11730831	-	Test Fixture
12599166	-	Mount, Telescope and Quadrant: M187
12984689	-	Mount, Telescope and Quadrant: M187A1

(Copies of these drawings may be requested online at Drawing-Request@pica.army.mil or from U.S. Army ARDEC, ATTN: AMSRD-AAR-AIS-TD, Picatinny, NJ 07806-5000)

2.3 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue of these documents are those cited in the solicitation or contract

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INTERNATIONAL ORGANIZATION FOR STANDARDIZATION

ISO 9000 - Quality Management and Quality Assurance Vocabulary

(Application for copies should be addressed to ANSI New York City Office Operations, 25 West 43rd Street, 4 floor, New York, NY 10036 or go online to http://www.ansi.org)

DEPARTMENT OF DEFENSE, DEFENSE SUPPLY AGENCY

DSAM 4145.8 - Radioactive Quantities in the DOD Supply System.

(Application for copies should be addressed to the Superintendent of documents, Government Printing Office, Washington, DC 20402.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.1 Regulations pertaining to radioactive devices. Marking, identification, handling and shipping of radioactive devices shall be in accordance with MIL-STD-129 and DSAM 4145.8 as applicable.

3.2 Materials. Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.3 Fabrication. The Mount shall be manufactured in accordance with Drawing F12599166 or F12984689 as applicable and drawings pertaining thereto.

3.4 General specifications. The contractor shall be responsible for the adherence to the requirements of MIL-F-13926.

3.5 Environmental.

3.5.1 Operating temperature. The mount shall meet the applicable requirements of 3.6.3 through 3.6.6.1 inclusive while exposed and thermally stabilized at ambient temperatures of +145°F (+63°C) ± 5°F (±3°C) and -50 °F (-46°C) ± 5°F (± 3°C) for a period of 4 hours ± 15 minutes at each temperature. Upon return to standard ambient temperature, +60 °F to +90°F (+16°C to + 32°C), from each operating temperature, the mount shall meet the requirements of this specification.

3.5.2 Storage temperature. The mount shall show no evidence of physical failure

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when thermally stabilized at ambient temperatures of +160 °F (+71 °C) ± 5°F (±3 °C) and -60 °F (-51 °C) ± 5 °F (± 3 °C) for a period of 4 hours ± 15 minutes at each temperature.

3.5.3 Shock. The mount shall withstand a total of 18 shock impulses, three in each direction of three mutually perpendicular axis. Each shock impulse shall be a half-sine wave with a time duration of 3 ±1 milliseconds. The peak amplitude of each shock impulse shall be 100 g's. Subsequent to shock, the mount shall not have been damaged or adversely affected and shall meet the requirements of 3.6 (see 4.6.1.2).

3.5.4 Vibration.

3.5.4.1 Vibration "A". Where required, the Mount shall withstand a total of 270 minutes ± 5 minutes of sweep-cycle vibration. The vibration shall be applied for 90 minutes ± 2 minutes along each of the three mutually perpendicular major axis. A complete sweep-cycle shall consist of vibration from origin, (5 hz at 1 inch double amplitude) to mid-point (5 g's ± 0.5 g's at 500 hz) to origin, and shall have a duration of 15 minutes ± 1 minute. Double amplitude shall be constant at 1 inch between 5 hz and 10 hz, and varied with frequency to maintain a constant 5 g's ± 0.5 g's acceleration between 10 hz and 500 hz. Upon completion of vibration, the mount shall exhibit no evidence of damage and shall meet all requirements of this specification.

3.5.4.2 Vibration "B". Subsequent to meeting the requirements of 3.6.1 the mount shall be vibrated in a vertical plane for a total of 15 minutes +/- 1 minute of sweep cycle vibration. A complete sweep cycle shall consist of vibration from origin (5 hz at 1 inch double amplitude) to mid-point (5 g's +/- 0.5 g's at 500 hz) to origin and shall have a duration of 15 minutes +/- 1 minute. Double amplitude shall be constant at 1 inch between 5 hz and 10 hz and varied with frequency to maintain a constant 5 g's +/- 0.5 g's acceleration between 10 hz and 500 hz. Upon completion of vibration, the mount shall exhibit no evidence of damage or physical failure and shall meet the requirements of this specification.

3.5.5 External tritium contamination. Tritium shall not be present on the external surfaces of the M187 mount in excess of 1000 dpm/100 sq.cm.

3.5.6 Sealing (counter box area). The counter box area of the mount shall be capable of sustaining an internal pressure of 2.0 ± .2 psi for a period of 2 hours (minimum) without loss of pressure in excess of 0.2 psi.

3.5.7 Purging. The counter box area of the mount shall be purged with dry nitrogen supplied at a pressure of 5 ± .2 psi until the dew point of the emergent gas is no higher than -25 degrees F. Purging apertures shall be quickly and positively closed to minimize intermingling of external atmosphere upon disconnecting of purging equipment.

3.5.8 Rain test. The Mount shall meet the requirements of Procedure I of the rain test of MIL-F-13926.

3.6 Performance.

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3.6.1 Orientation. The mount shall be positioned and orientated as follows:

a. The mounting surface (internal surface) of the mount shall be in a vertical plane.

b. The mount locating keyway shall be horizontal within 10 seconds.

c. The top locating surface for the telescope, shall be parallel to the mount's locating keyway within 0.5 mil.

d. The elevation and cross level vial bubble shall be centered.

e. The gunner's quadrant pads shall be level in reference to location keyway within 0.2 mil.

f. The locating key for the telescope shall be parallel to the vertical mounting surface within 1.0 mil.

3.6.2 Movement.

3.6.2.1 Elevation travel. The Mount shall travel through an excursion of at least 1333 mils in elevation and 89 mils in depression from the position established in 3.6.1 when the correction counter is set at zero.

3.6.2.2 Correction counter. The correction counter shall travel through an excursion of plus and minus 9 mils.

3.6.2.3 Cross level travel. The cross level knob shall move the mount through a minimum angle of 20 degrees to the right and 20 degrees to the left from the position established in 3.6.1.

3.6.3 Mount rigidity. With the mount positioned as in 3.6.1 apply a 20 pound load at the right side of the telescope mounting flange in line with the locating pins. Release the load and record the movement. Apply the same load at the left side of the telescope mounting flange in line with the locating pins. Release the load and record the movement. The total movement as indicated by the two readings shall not exceed the values of Table I.

TABLE I
MOVEMENT RIGIDITY

ELEVATION (in mils)	CANT (in degrees)	TOTAL MOVEMENT (in mils)
0	0	.25
800	0	.75
1100	0	1.75
1333	0	3.50

3.6.4 Accuracy. With the mount positioned in accordance with 3.6.1, the following conditions shall be met.

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3.6.4.1 Plumb travel. With the cross level vial bubble centered, the line of sight shall not deviate from a plumb line by more than 0.5 mil when the elevation knob is actuated through its full range of movement.

3.6.4.2 Elevation error. Elevation error shall not exceed 0.5 mil excluding backlash as read in 200 mil increments between zero and 1200mils.

3.6.4.3 Backlash.

3.6.4.3.1 Elevation mechanism. Backlash in the elevation mechanism shall not exceed 0.5 mil.

3.6.4.3.2 Cross level mechanism. Backlash in the cross level mechanism shall not exceed 0.5 mil.

3.6.4.4 Correction counter. The correction counter shall be checked at plus and minus readings of 5 and 9 mils. The angle of the Gunner's Quadrant seat shall correspond to the correction values within 0.2 mil when the elevation counter is reset to zero reading.

3.6.4.5 Counter alignment. Variation in the alignment of the numbers in the counter shall not exceed one sixteenth inch (1/16").

3.6.4.6 Correction counter operation. Plus values of 5 and 9 mils put on the correction counter shall subtract an equal number of mils from the elevation counter. Minus values of 5 and 9 mils put on the correction counter shall add an equal number of mils to the elevation counter.

3.6.5 Operation torques. The maximum of operating torques shall not exceed the values in Table II at temperatures indicated therein.

TABLE II

	AT STANDARD AMBIENT TEMPERATURE OF +60 ºF TO +90 ºF AND AT +150ºF		AT -50ºF
Cross Level Knob	4 lb-inch min	12 lb-inch max	25 lb-inch max
Elevation Knob	4 lb-inch min	14 lb-inch max	25 lb-inch max
Correction Knob Elevation	2 lb-inch min	4 lb-inch max	8 lb-inch max

3.6.6 Correction.

3.6.6.1 Cant correction. The mount shall make left and right cant correction in accordance with the values in TABLE III. The elevation counter shall read within plus or minus 0.5 mils of the elevation angle of the quadrant seat.

TABLE III

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CANT CORRECTION

QUADRANT Elevation (in mils)	CANT ANGLE (in mils)	CORRECTION (in mils)	TOLERANCE (in mils)
500	177.8	96.1	0.5
800	177.8	180.5	1.0
1000	177.8	271.9	2.0

3.7 Illumination. The level vial bubble graduation lines and two counters shall be clearly distinguishable when observed in ambient light conditions ranging from dusk into darkness with and without the use of a 50% neutral density filter.

3.8 Interchangeability. Interchangeability shall be verified by the insertion, seating, securing and subsequent removal of maximum fit gage on the mount's top surface.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in ISO 8402.

4.2 First article (initial production) approval. The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval of test shall consist of three (3) mounts plus three (3) each of all items covered by Supplementary Quality Assurance Provisions (SQAP). The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the government, shall inspect the sample to insure that it conforms to all requirements of the contract and submit a record of this inspection, with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing. When the contractor is responsible for conducting first

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article approval test, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for apporval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product. Unless otherwise specifies by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-1916.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies. All components and subassemblies shall be inspected in accordance with the inspection provisions contained in SQAP listed in the Technical Data Package (TDP). In the absence of SQAP's the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2.2 Final acceptance inspection. Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristics) basis in accordance with MIL-STD-1916 and the sampling plans specified in Table IV herein. Examination and tests for Packaging, Packing and Marking shall be in accordance with MIL-STD-2073-1 and Section 5 herein. The tabulated classification of defects in Table IV and V shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE IV – CLASSIFICATION OF DEFECTS

NO	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
MAJOR: 100% INSPECTION			
101	Sealing	3.5.6	4.6.1.6
102	Purging	3.5.7	4.6.1.7
103	Orientation	3.6.1	4.6.2.1
104	Illumination	3.7	4.6.3
105	Interchangeability	3.8	4.6.4
MAJOR: AQL 0.65% DEFECTIVE			
106	Vibration "B"	3.5.4.2	4.6.1.4
107	External Tritium Contamination	3.5.5	4.6.1.5
108	Elevation Travel	3.6.2.1	4.6.2.2
109	Correction Counter	3.6.2.2	4.6.2.3
110	Cross Level Travel	3.6.2.3	4.6.2.4
111	Mount Rigidity	3.6.3	4.6.2.5

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112	Plumb Travel	3.6.4.1	4.6.2.6
113	Elevation Error	3.6.4.2	4.6.2.7
114	Elevation Backlash	3.6.4.3.1	4.6.2.7
115	Cross Level Backlash	3.6.4.3.2	4.6.2.4
116	Correction Counter	3.6.4.4	4.6.2.8
117	Counter Alignment	3.6.4.5	4.6.2.9
118	Correction Counter Oper	3.6.4.6	4.6.2.10
119	Operating Torques	3.6.5	4.6.2.11
120	Cant Correction	3.6.6.1	4.6.2.12

MINOR: NONE

4.3.3 Acceptance and rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejections" as specified in MIL-STD-1916.

4.4 Special sampling.

4.4.1 General. Three telescopes shall be selected at random as a special sampling from each 100 produced or one months production whichever occur first. The sample shall meet the requirements of Table V.

TABLE V

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301	Fabrication	3.3	Applicable drawings – visual
302	General Specification	3.4	MIL-F-13926
303	Operating Temperature	3.5.1	4.6.1.1
304	Storage Temperature	3.5.2	4.6.1.1
305	Shock	3.5.3	4.6.1.2
306	Rain	3.5.8	4.6.1.8

4.4.2 Failure of sample. Should any one of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.3).

4.5 Test equipments.

4.5.1 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.2 Government furnished inspection equipment. Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

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4.5.3 Contractor furnished inspection equipment.

4.5.3.1 Government design. Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the list of inspection documents when provided with the TDP.

4.5.3.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. This, concept, construction, materials, dimensions and tolerance limit used in the design of the test equipment shall be so selected and controlled as to insure the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.3.3 Test equipment. In conjunction with 4.5.3.3, the following standard test equipment shall be utilized in the performance of the applicable test as specified in 4.6.

Nomenclature	Description
1. Hot and cold chamber	Standard type conforming to the accuracies specified under Test Facilities of MIL-F-13926.
2. Light Source Standard	
3. Torque wrench	Standard shop type accurate to 10 % of the drawing tolerance.

4.5.3.4 Device to test performance. Test equipment used to perform acceptance shall be equal to or exceed the accuracies depicted on drawing 11730831. Any equipment used to perform the tests shall be submitted to the government for review (see 6.3).

4.6 Examinations and tests.

4.6.1 Environmental.

4.6.1.1 Storage and operating temperatures. The mount shall meet the applicable requirements of 3.6.3 to 3.6.6.1 inclusive while exposed and thermally stabilized at ambient temperatures of +60 degrees F to +90 degrees F. The mount shall then be placed in a government approved test chamber and the temperature stabilized at +150 degrees F

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and meet the requirement of 3.6.5. The temperature shall then be stabilized at +160 degrees F for a period of 6 hours. The mount shall then be thermally stabilized at a temperature of -50 degrees F and meet the requirements of 3.6.5. The mount shall then be stabilized at an ambient temperature of -50 degrees F for a period of 6 hours. Subsequent to the temperature cycle, the mount shall be visually and examined and shall show no evidence of physical failure and meet all requirements of this specification at standard ambient temperature (+60 degrees to +90 degrees F).

4.6.1.2 Shock. The shock-producing apparatus shall be capable of producing the shock pulse and shall be calibrated for conformance with the requirement as outlined in 3.5.3. The frequency response of the complete measuring system, from any accelerometer through the readout instrument, shall be accurate to within ten percent of the specified tolerance. Subsequent to the 18 shock impulses the mount shall be visually examined for damage and shall meet all requirements of this specification.

4.6.1.3 Vibration "A". This test is applicable to the First Article sample only. The vibration exciter or transition table shall be approved by the government. The mount shall be attached by its normal means, either directly to the vibration exciter or transition table, or by means of a rigid fixture capable of transmitting the vibration conditions. Precautions shall be taken in the establishment of mechanical interfaces to minimize the introduction of extraneous responses in the test setup. The test load shall be distributed as uniformly as possible on the vibration exciter table in order to minimize the effects of unbalanced loads. The input control sensing device(s) shall be rigidly attached to the vibration table or fixture if used, as near as possible to the attachment point of the mount. The vibration shall applied as specified in 3.5.4.1. Upon completion of the vibration test, the mount shall meet the requirements of this specification.

4.6.1.4 Vibration "B". This test is to be performed as outlined in Procedure V of MIL-F-13926, however the frequency and duration shall be as outlined in 3.5.4.2. At the conclusion of the test the mount shall meet the requirements of this specification.

4.6.1.5 External tritium contamination. Contamination test shall be performed by wiping the M187 telescope with filter paper moistened in distilled water. The wiping shall be performed with moderate finger pressure. The damp filter paper shall be placed in a container with the proper portion of scintillation liquid. The read out shall be within 10% of the known standard value when counted to a total of 2000 disintegrations. The actual test for determining contamination is performed by placing the container into the scintillation system. The measured contamination shall be within the limits specified in 3.5.5.

4.6.1.6 Sealing (counter box area). The counter box portion of the mount shall be pressurized internally to the psi value specified in 3.5.6 with dry nitrogen for the length of time specified in 3.5.6. Dry nitrogen used in this test shall have a dew point of at least -25 degrees F. Accuracy of the test equipment used on this test shall be equal to or exceed the accuracies depicted on Drawing F8565556.

4.6.1.7 Purging. The counter box area of the mount shall be flushed with dry

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nitrogen gas to conform to the requirements of 3.5.7. A dew point tester (standard shop type test equipment) shall be used to measure the nitrogen dew point for conformance with 3.5.7. Purging to be performed at standard ambient temperature and prior to performing temperature test. Pressure of nitrogen remaining within the counter box area to be between .25 and .5 psi.

4.6.1.8 Rain. The mount shall be subjected to the rain test outlined in Procedure I of MIL-F-13926. After completion of the rain test, a visual examination shall be made to determine leakage.

4.6.2 Performance testing.

4.6.2.1 Orientation. Using test equipment as described in 4.5.3.4, the mount shall be set up and checked for conformance with 3.6.1.

4.6.2.2 Elevation travel. With the mount and fixture positioned as outlined in 4.6.2.1, rotate the mount's elevation knob to reflect maximum elevation. Note the reading on the elevation counter. Set this reading into Inspection Aid (Gunner's Quadrant) A8658940. Position the inspection aid on the quadrant seat assembly and by means of the fixture elevation handwheel, relevel the inspection aid. Observe the elevation level bubble of the mount with respect to the quadrant level vial. Both must indicate level within the limits outlined in 3.6.2.1. Repeat this procedure in the opposite direction. Any non-compatibility in level vial bubble position shall be measured by means of the quadrant.

4.6.2.3 Correction counter. Rotate the correction counter knob to indicate that the maximum correction range is within the limits of 3.4.2.2.

4.6.2.4 Cross level travel and backlash. With the mount positioned on the fixture as outlined in 4.6.2.1 and with the auxiliary telescope's sighting device assembled to the mount, rotate the cant mechanism to the full extent of travel. Place the inspection aid over the appropriate set of pads of the sighting device and read the cant angle. This reading must be equal to or exceed the minimum requirement outlined in 3.6.2.3. Repeat this procedure at the opposite extreme setting. Return the cross leveling mechanism somewhere between zero and maximum angle and scribe a line on the casting and cross leveling knob and obtain a cross level angle reading using the inspection aid, record this reading. Continue to rotate the cross leveling knob one revolution beyond the index mark and return to the index mark. Exercise care not to override this point of coincidence and record this reading using the inspection aid. The difference, if any, is considered backlash and must conform to the limits outlined in 3.6.4.3.2.

4.6.2.5 Mount rigidity. With the fixture positioned as outlined in 4.5.2.3, position the mount as specified in 3.6.1 with the level vial bubbles centered. Check rigidity by gradually applying and releasing the pressure specified in 3.6.3 at the top left side of the telescope support as specified in 3.6.3. Take the reading on the test target and record the movement. Repeat this procedure gradually applying and releasing the specified pressure at the top right side of the telescope support. Take the reading on the test target and record the movement. The total movement shall be within the allowable maximum deviation

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specified in Table I of 3.6.3. The same procedure shall be followed as the simulated gun trunnion (cross leveling fixture) is moved from the "O" position to the 1333 mil elevation position to conform to the requirements of Table I. The maximum movement at the various settings shall not exceed the mil values specified in Table I.

4.6.2.6 Plumb travel. With this mount and fixture set as outlined in 4.6.2.1, this test is made by sighting through viewing telescope F5549108 and observing coincidence between the telescope's optical axis and the intersecting point of target centerlines while simultaneously rotating the mount in elevation and the test fixture an equal amount in the opposite direction. The optical line of sight shall not deviate from the intersecting point by more than the limits outlined in 3.6.4.1 as measured by target mil blocks. During this total travel the cross level vial shall remain level within the limits of 3.6.4.1.

4.6.2.7 Elevation error and backlash. With the fixture and mount positioned as outlined in 4.6.2.1 and with the auxiliary telescope's sighting device assembled to the mount, insert an elevation into the mount from low to high thus excluding backlash. Rotate the fixture handwheel until the mount elevation level vial bubble is centered. Using a Gunner's Quadrant (inspection aid) on the fixture quadrant seat assembly, measure the actual elevation. The difference between the whole number, mount elevation and the actual reading from the Gunner's Quadrant shall not exceed the limit outlined in 3.6.4.2. Place the Gunner's Quadrant on the auxiliary telescope's sighting device and center the quadrant level vial bubble. Record this Gunner's Quadrant reading. Rotate the mount elevation knob approximately one revolution beyond the elevation. Rotate the knob in the reverse direction back to the original whole number elevation, making sure not to override this position. Center the Gunner's Quadrant level bubble. Record this Gunner's Quadrant reading. The difference between the first Gunner's Quadrant reading and the second is considered backlash and shall not exceed the limit outlined in 3.6.4.3.1.

4.6.2.8 Correction counter. With the mount and fixture set as outlined in 4.6.2.1, rotate the correction counter to the minimum setting specified in 3.6.4.4. Return the elevation counter to zero. By means of the fixture, level the elevation level vial of the mount. Do not override this level position. Place the inspection aid on the quadrant seat assembly and set the inspection aid vial to indicate level. Read the angle on the inspection aid. This reading to be identical to the correction counter reading within the limits of 3.6.4.4. Repeat this procedure at the setting specified in 3.6.4.4 each side of zero correction.

4.6.2.9 Counter alignment. Actuate the elevation knob through its complete travel range. During this travel make a minimum of 2 stops each side of zero elevation, and by standard measuring equipment, determine if digit alignment is within the limits outlined in 3.6.4.5.

4.6.2.10 Correction counter operation. This test may be performed in conjunction with the previous test. Visually determine that when a negative correction is inserted in the mount, the elevation counter reading increases an equal amount. Also, determine that when a positive correction is inserted in the mount, the elevation counter reading decreases an equal amount. Using the test procedure outlined in 4.6.2.1, determine that the inserted

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correction factors are recorded on the elevation counter as specified in 3.6.4.6.

4.6.2.11 Torques. All torque readings specified in 3.6.5 shall be measured using standard torque wrench and appropriate adapters. All readings to conform to the values specified in 3.6.5 for the temperature specified.

4.6.2.12 Cant operation. With the mount and fixture positioned as outlined in 4.6.2.1, rotate the auxiliary telescope's line of sight 1600 mils and engage the locating pin in the proper locating hole indicated in the set up instructions similar to 11730831. The line of sight shall coincide with the vertical target line with the limits outlined in 3.6.6.1. Elevation angle of the quadrant seat shall coincide with the elevation reading shown on the elevation counter within the limits of 3.6.6.1. Elevate and cant the fixture to selected set of values (Table III) and set the mount off an equal amount in the opposite direction. The telescope's line of sight shall coincide with the target vertical centerline within the limits outlined in 3.6.6.1.

4.6.3 Illumination. Place 50% neutral density filter in front of the counters and level vials. In a darkened area visually inspect for conformance with 3.7.

4.6.4 Interchangeability. Each mount shall be inspected for maximum fit conditions using the interchangeability gage 11747889. Any mount which does not accept the interchangeability gage shall rejected.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. Packaging, packing, and marking shall be in accordance with MIL-STD-2073-1, MIL-STD-129 and applicable packaging data sheets. The level of protection shall be specified in the procurement document.

6. NOTES

6.1 Intended use.

6.1.1 M187 and M187A1. The Telescope Mount and Quadrant: M187 and M187A1 is primarily intended to support a panoramic telescope on the Light Howitzer, M119A1 and M119 Series. The mount provides cant correction and elevation adjustments for the telescope.

6.2 Ordering data. Procurement data should specify:

a. Title, date and number of this specification.

b. Serial numbers to be used for identification marking.

c. Applicable packaging data sheet number.

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d. Selection of an applicable level of preservation and packaging.

e. Requirements for first article testing.

f. Contract data requirements for submission of inspection equipment designs conforming to Data Item Description DI-R-1714.

6.3 Inspection equipment design.

6.3.1 Submission of designs for approval. Contractor designs for final acceptance inspection must be approved by the Government prior to fabrication or procuring the equipment. The contractor is referred to MIL-HDBK-204 for guidance. Submission of design concept on inspection equipment is permissible for tentative approval. The completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. Submit designs as required to: Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAF-I (D), Picatinny Arsenal, NJ 07806-5000. When the contractor submits inspection equipment designs to the government for approval, he must give the following information in his letter of transmittal:

a. The contract number.
b. The contract item (name, model number, etc).
c. The designs remaining to be submitted and the expected date of submittal.

6.4 Subject term (key word) listing.

Mount
Fire Control
Telescope
Radiation
Quadrant

6.5 Hazardous material.

6.5.1 Tritium. The M187 Mount, Telescope and Quadrant contains tritium, a radioactive material. Prospective bidders should be cognizant of the need for a Nuclear Regulatory Commission License governing the procurement, processing, handling and shipping of radioactive material.

6.5.2 Lithium Batteries. The M187A1 utilizes Lithium batteries. Prospective Bidders must be cognizant of the need to handle and/or dispose of these batteries in accordance with all Local, State and Federal Laws.

6.6 Packaging data sheet drawings. The following packaging data sheet drawings should be obtained from U.S. Army ARDEC, ATTN: AMSRD-AAR-AIL-P B455, Picatinny, NJ 07806-5000

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12599166	-	Packaging of Mount Telescope and Quadrant: M187
12984689	-	Packaging of Mount, Telescope and Quadrant: M187A1

6.7 Amendment notations. The margins of this specification are marked with vertical lines to indicate where modifications from this amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodian: Army-AR

Preparing activity: Army-AR
(Project 1240-2006-004)

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